

PROJECT PROPOSAL

Project Title: UrbanMend - An AI-Powered Crowd-Sourced System for Public Issue Prioritization

Research Area: Artificial Intelligence, Web Development, and Civic Technology

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1. Introduction

Background

In rapidly growing urban environments, maintaining public infrastructure such as roads, streetlights, and waste management systems is a constant challenge for local authorities. Currently, many citizens feel disconnected from the maintenance process, often finding it difficult to report issues or track their resolution. **UrbanMend** is designed to bridge this gap by providing a centralized, intelligent platform that empowers citizens to act as the "eyes and ears" of the city. By integrating AI, the system moves beyond simple data collection to providing actionable insights for smarter city management.

Problem Statement

Existing public reporting methods are often fragmented, relying on manual phone calls, emails, or outdated web forms that lack transparency and real-time updates. Furthermore, municipal authorities are often overwhelmed by the volume of reports and lack a standardized way to identify which issues are the most urgent. This results in delayed response times for critical problems (like dangerous potholes or broken streetlights in high-crime areas) while less urgent issues might be addressed first.

Objectives

- To develop a user-friendly web platform for citizens to report public infrastructure issues with geolocation data and image support.
- To implement AI/ML logic that automatically categorizes reports (e.g., Sanitation, Infrastructure, Utilities) and assigns a priority score based on severity and location.
- To provide authorities with a centralized dashboard to track, manage, and update the status of reported issues in real-time.
- To improve the transparency of public works by allowing citizens to track the progress of their reports.

Scope

- **Included:** Development of a responsive web application, a backend AI classification engine (NLP/Keyword analysis), a database for report storage, and an administrative dashboard for government officials.
- **Excluded:** Mobile native application development (focus is on a mobile-responsive web version), direct integration with government financial/budgeting systems, and physical repair of infrastructure.

2. Requirements Analysis

Functional Requirements

The **UrbanMend** system will deliver the following core functionalities:

1. **User Authentication:** Secure registration and login for citizens; specialized verification for administrative/authority accounts.
2. **Smart Report Submission:** A form allowing users to describe the issue, upload a photo, and select a location via a map interface (or auto-detect GPS).
3. **AI Classification Engine:** A backend module that uses Natural Language Processing (NLP) to read report descriptions and categorize them automatically.
4. **Priority Scoring Algorithm:** An automated logic that calculates urgency based on:
 - **Keywords:** (e.g., "danger," "accident," "flood" increase priority).
 - **Frequency:** Multiple reports of the same issue in one location increase the priority.
 - **Proximity:** High priority for issues near hospitals, schools, or main highways.
5. **Authority Dashboard:** A filtered view for officials to see high-priority "hotspots" and update status (e.g., "Pending" to "Resolved").
6. **Notification System:** Automated email or system alerts to users when the status of their report changes.

3. Expected Outcomes

Deliverables

- A fully functional web application prototype.
- A technical report detailing the AI/ML model's logic and accuracy.
- A user manual for both citizens and administrative authorities.

Impact and Improvements

This system will significantly reduce the manual workload for public works departments by automating the initial triage of reports. By providing authorities with a "heat map" of prioritized issues, resources can be allocated more effectively, leading to faster response times and higher citizen satisfaction with public services.

4. Conclusion

The **UrbanMend** project represents a modern, AI-driven approach to civic engagement and urban maintenance. By leveraging crowd-sourced data and intelligent prioritization, the system ensures that public issues are not just reported, but managed efficiently based on actual need. This project demonstrates how technology can be used to build more resilient, responsive, and "smarter" communities.

5. References

Similar Websites & Platforms

1. **FixMyStreet (mySociety)** – <https://www.fixmystreet.com>: An open-source platform in the UK that allows citizens to report street issues directly to local councils.
2. **SeeClickFix (CivicPlus)** – <https://seeclickfix.com>: A global platform used by municipalities to empower citizens to report non-emergency neighborhood issues.
3. **CitySourced** – <https://www.citysourced.com>: A real-time mobile civic engagement platform for identifying and reporting public safety and environmental issues.

Academic & Technical Sources

4. **Aykul, O., et al. (2025).** *Proactive Complaint Management in Public Sector Informatics Using AI*. MDPI Applied Sciences. (Discusses AI-based complaint clustering).
5. **Pak, B., & Chua, A. (2017).** *FixMyStreet Brussels: Socio-Demographic Inequality in Crowdsourced Civic Participation*. Journal of Urban Technology.
6. **Deshmukh, S., et al. (2025).** *Pothole Complaint and Tracking Platform*. International Journal of Advanced Research in Science, Communication and Technology (IJARSCT).